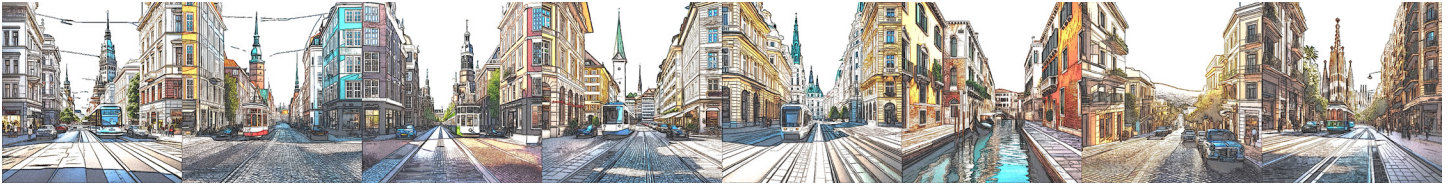


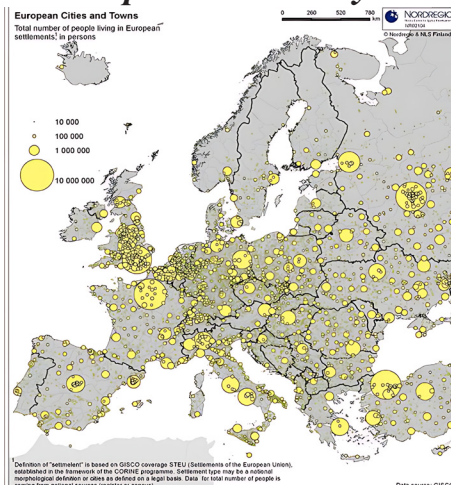
Examining the Factors for Sustainable Urban Development in Europe: *Balance of Economy, Social Equity, and Environmental Sustainability.*



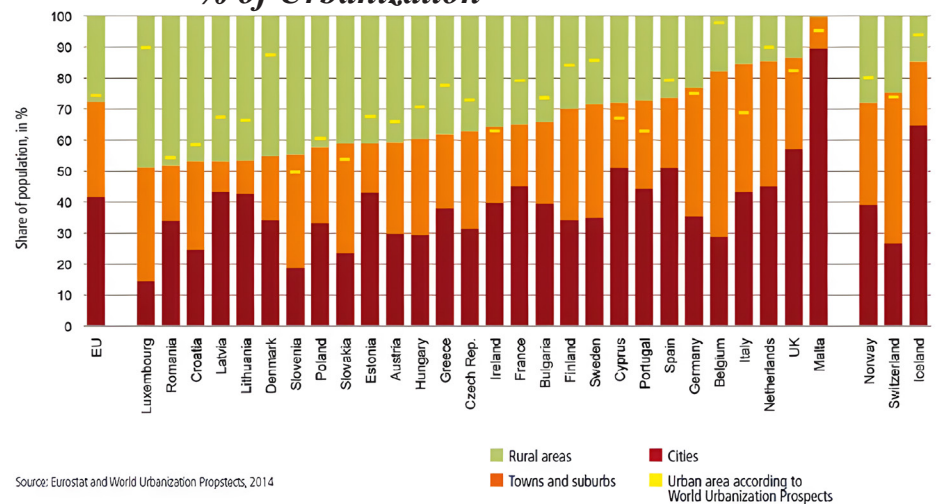
Ethan Fryer

The Rapid Urbanization of European cities in recent decades has generated a diverse population of urban challenges and opportunities. With the Influx in Populations growing 35% in the last 70 years and Societal demands evolving, the decision of urban planning becomes increasingly Vital. From Barcelona's vibrant streets to Prague's historic cityscape, European cities stand as testaments to the intricate dance between urban development and preserving cultural, economic, and environmental values.

Population density



% of Urbanization



In the past five decades, Europe has shifted from being a primarily rural continent to a more metropolitan one. This societal pivot has pushed cities to become hubs for economic transactions, cultural interactions, and communal experiences. Europe's cities have adjusted to the evolving European Union and positioned themselves within a more integrated continental framework. With the world entering the age of metropolitan Urbanization, Europe's cities have become a critical platform for action and innovation.

Through studying abroad, I formulated the question: "What are the key elements of successful and sustainable urban development policies in European cities?" I came upon this question in Copenhagen after recently visiting the second-largest city in Denmark, Aarhus. I noticed the regional differences in the urbanization between the two cities. Copenhagen, being the capital city, has an overwhelming sense of innovative industrialization with a subtle tendency toward cultural influence. In contrast, Aarhus felt almost the opposite, with a vast cultural influence focused on preservation and subtle integrated urban growth.

These subtle differences in urban evolution only became more noticeable as the trip went on, leaving me wondering, "How can urban planning effectively balance economic growth, social equity, and environmental sustainability to create resilient and inclusive urban environments?" by the end of the trip, I was able to feel the vast Differences in urban development, through each European region. From Venice's Tourism industry to Barcelonas's Hatred towards Tourists, it was clear that European cities have required different types of investment to adapt to distinctive and different urban futures.

A resilient and inclusive urban environment is a city that can withstand challenges, adapt to changes, and prioritize equal access and opportunities for all its residents, promoting sustainability, community well-being, and social equity. EU Government literature states, “European cities are adopting an integrated approach to sustainable urban development, guided by the three pillars of sustainability: smart growth, green growth, and inclusive growth.” One may misunderstand the word sustainability, which is often assumed to mean just environmental protection. It was not till 1987 that the three pillars of sustainability Published in “Our Common Future” Gave the word a complex concept with three core pillars: environmental, social, and economic. These pillars provide a framework for evaluating sustainability for organizations, products, services, and technologies. Today, these pillars represent the most popular model for evaluating and understanding sustainable development. Why is this important? The EU Framework for Action for Sustainable Urban Development highlights the importance of better coordinated and targeted community action for urban problems, organized under four interdependent policy aims: economic prosperity and employment, social inclusion, and environmental protection and improvement.

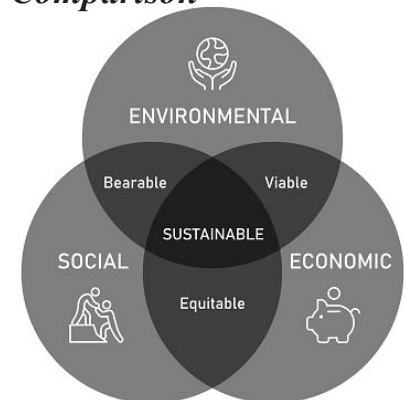
European cities increasingly focus on building resilience and enhancing environmental sustainability, promoting resource efficiency, reducing pollution, and addressing climate change. The presence of accessible, high-quality green spaces in cities supports physical and mental health and social well-being. Broken down in the Urban Sustainability Topic published by the European Environment Agency is why this is so important, “About three-quarters of Europeans live in cities, according to UN-Habitat. With their concentrated population and infrastructure, cities are especially vulnerable to certain effects of climate change and other environmental stresses like heatwaves, flash floods, pollution, noise, and soil contamination.” The necessity to build sustainability is at an all-time high. “Building stock accounts for 40% of energy consumption and 36% of GHG emissions. Currently, 75% of the European building stock is energy inefficient.” These challenges have a disproportionate impact on vulnerable portions of the population. The Paper further discusses how the “Cities have their governments and are often best placed for local action to tackle and resolve environmental challenges.” These EU policies aim to enhance cities’ resilience and capacity to transition towards sustainability, addressing environmental, social, economic, and cultural issues.

Europe’s Choice to Tackle the issue on the local scale prioritizes community engagement and inclusive growth. This involves active participation of the community in the planning and decision-making process to ensure that their needs and priorities are met. Inclusive growth aims to ensure that all members of society share the benefits of development, emphasizing social inclusion, equal access to opportunities, and the reduction of inequality. European cities are working towards providing affordable housing, quality education, healthcare, and social services for all residents while promoting economic competitiveness and high-quality jobs.

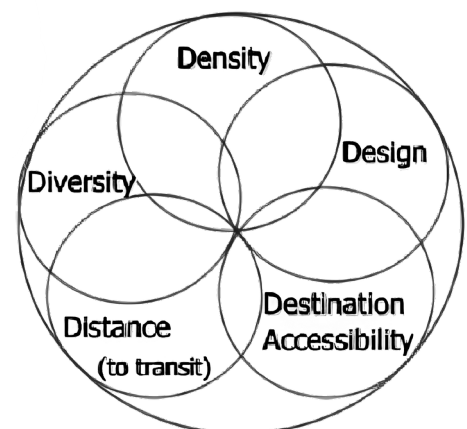
3 pillars of Sustainability



3 pillars of Sustainability Comparison



5 D's of the Built Environment



European urban planning and government entities emphasize the intertwining principles of economy, social equity, and environmental sustainability as fundamental to successful urban development.

Economic Considerations:

They are creating strong local economies through diverse land use, sustainable practices, efficient transportation, and partnerships between the public and private sectors while following regulations and considering social and environmental factors for well-rounded and sustainable development.

Social Equity Prioritization:

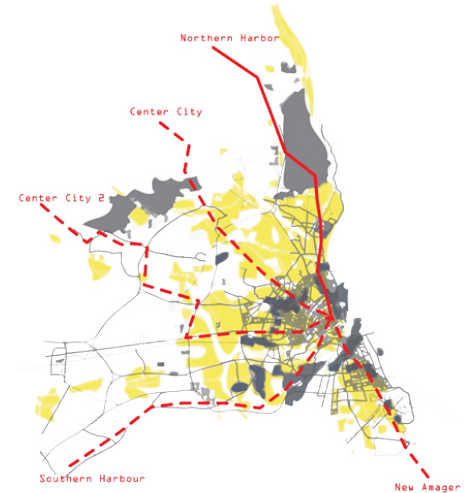
Priority is given to social equity by ensuring diverse and affordable housing options, engaging communities in decision-making, providing equal access to services, fostering a social mix in neighborhoods, and promoting inclusivity for the well-being of all residents.

Environmental Sustainability Integration:

A focus on integrating environmental sustainability by incorporating green spaces, promoting energy efficiency in buildings and infrastructure, implementing effective waste management, and adapting plans to address climate change impacts, all aimed at creating resilient and eco-friendly urban environments.

By analyzing economic factors such as Barcelona's Smart City Initiatives, Exploring social equity through Amsterdam's Bicycle-Friendly Infrastructure, and most importantly, environmental sustainability with Helsinki's Green Infrastructure, we will have a better understanding of how to create resilient and inclusive urban environments through the eye of a European urban planner.

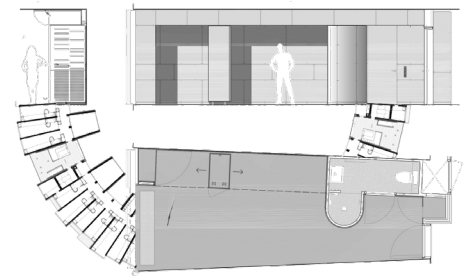
Zoning & Transportation



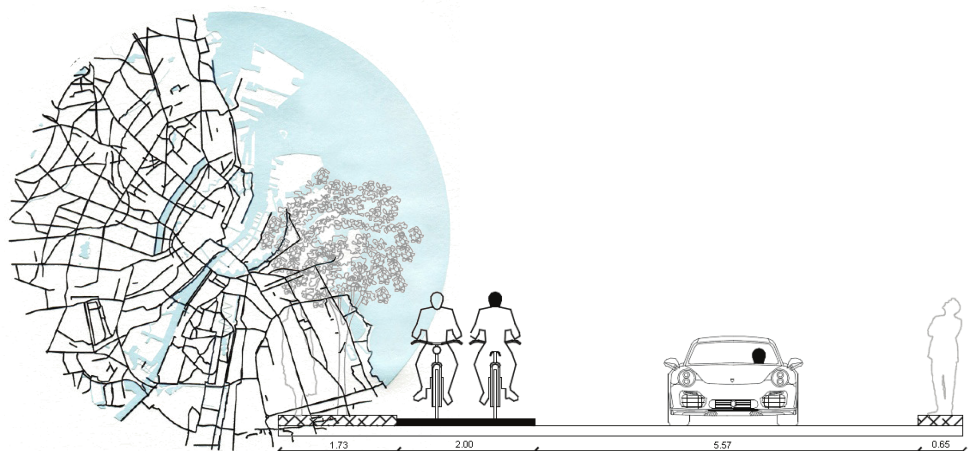
Linear City Plan Copenhagen 2078

Old Copenhagen
Industrial Area
Zero Gravity Zones

Social Environments

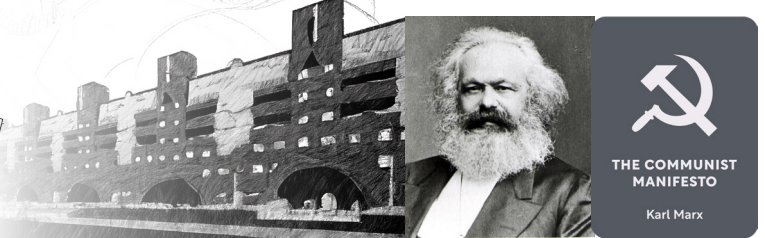
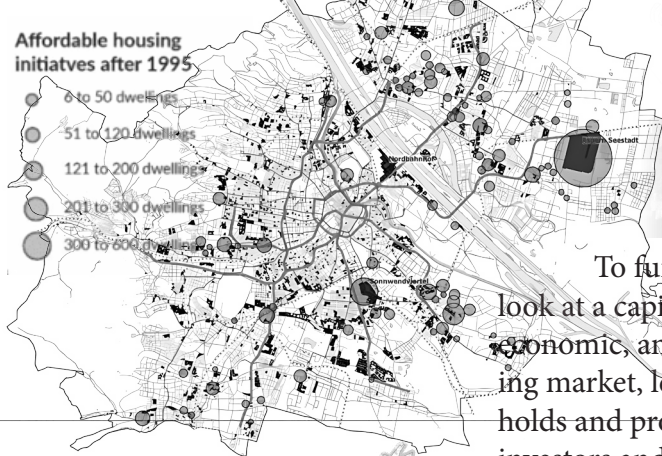


Human Scale



Economic factors include land use and property values, infrastructure investment, mixed-use development, and affordable housing. These factors are crucial in shaping planning and architecture, influencing cities' design, development, and sustainability.

The most significant economic factor in Urban planning is Economic Growth. A Superior example of economic growth is Vienna's Affordable Housing Model. Vienna, Austria's capital, has a storied past as an imperial city, hosting both the Holy Roman Empire and the Austro-Hungarian Empire. Renowned for its quality of living, Vienna houses 7% of Austria's population and a third of its full-time students. With more than 60 percent of the city's residents living in social housing, Vienna's affordable housing model is a complex and innovative system celebrated for its effectiveness in providing superior, affordable housing to the city's residents. The city's commitment to providing high-quality housing for the working class dates to the "Red Vienna" period of the early 20th century and remains a priority. Much of Vienna's success in economic growth strategies comes from the Period Red Vienna when the government of Vienna, Austria, was led by socialists. During this time, they implemented progressive and inclusive policies, focusing on public housing, education, and social services. The term "Red Vienna" highlights the left-leaning or socialist nature of these initiatives. This period of socialist policies created a foundation for inclusive Economic growth. Investing in affordable housing and education gives more people better living conditions and education opportunities. This, in turn, contributed to a skilled and healthier workforce. A robust social safety net also helped reduce poverty and inequality, fostering a more stable and productive society. In 2018, 2019, 2022, and 2023, Vienna was the most livable city in the world, and it earned top marks for its stability, education, health care, and infrastructure.



To further understand the impact of affordable housing, one must look at a capitalistic affordable housing model like Berlin. Historical, economic, and institutional factors have influenced the city's rental housing market, leading to a lack of affordable housing for low-income households and protests of increasingly burdensome rents. The conflict between investors and renters has been a focal point, and the government's actions reflect a commitment to balancing growth and modernization with affordable housing. In response to the affordability challenge, Berlin's government has introduced measures to maintain rental affordability, including acquiring new housing units and developing an affordable housing plan that links rent to salary. The city's economy is centered on its technology and service sectors, which have experienced rising population growth, leading to concerns about the strain on the housing supply. Population growth, increased tourism, and gentrification have also influenced this growing demand. In 1945, at the end of World War II, 20 percent of the German housing stock was reduced to rubble, with more than 2 million homes destroyed and another 2 million left uninhabitable. After this period, the government established new housing laws, such as generous tax exemptions for public and private construction entities, the homebuilding industry flourished, and Germany halved its housing shortage by 1956. All this led to the significant Housing crisis we see in Berlin today; it was not until 2015 that the Berlin Senate declared that the city's renters, who make up 85 percent of Berlin's population, were paying too much for rent.



Many economic Factors are considered when discussing urban development, especially with various economic and governmental practices in Europe. One general truth stands true: the more affordable housing is, the better quality of life occupants in the city will have, leading to an overall healthier and more productive city. Cohousing is a relatively new affordable housing strategy in Berlin, Germany. It was co-founded by Michael LaFond, an American living in Berlin and the founder of the Institute for Creative Sustainability. According to Hud User's article on "Global Cities and Affordable Housing: Berlin," "For years, many young activists squatted in abandoned buildings throughout Berlin to save them from demolition and to live relatively cheaply or, in some instances, for nothing. The city found it too laborious to evict the activists, so it agreed with the activists to purchase the buildings from absentee owners with the understanding that it would refurbish them." The entire goal of a cohousing model is to bring a healthier and more productive opportunity to what would have been just an unusable economic money pit. Hud Users say that as the city's population grows, cohousing helps with affordability because it is less expensive to develop, and developers rarely, if ever, seek a profit.

If one were to try to determine the health of a city, one would look at the factors of social inequality that play a part in the everyday lives of its citizens. Social equity in urban planning refers to the fair and just distribution of resources, opportunities, and benefits within a city to ensure that all residents, regardless of their background or socio-economic status, have equal access to the benefits of urban development.



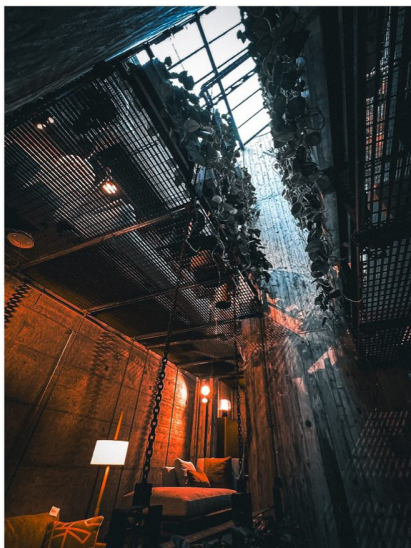
The heart of freedom in the European Union, known to many as the city where everyone is free to be themselves, is Amsterdam, renowned for its iconic canals and historic charm. Urban planning takes on a distinctive character beyond Storybook views, standing out as a beacon of social equity, where every neighborhood is thoughtfully designed to celebrate diversity, making Amsterdam a shining example of an inclusive and accessible urban haven. Amsterdam is home to 1.1 million residents from 180 nationalities, its population pyramid is more top-heavy, and its unemployment is a mere 3.8%. The city has recognized the importance of diversity and inclusion, focusing on ensuring that all residents can participate and that no one is excluded or made to feel like an outsider. Amsterdam's policy is focused on diversity and inclusion, emphasizing the concept of mutual responsibility based on the principle of 'equal rights, equal opportunities.' We see this attitude with Its affordable Housing initiatives and inclusive housing policies. Its mixed-income neighborhoods aim to prevent socio-economic segregation by providing affordable housing options throughout the city; this fosters the ability for residents from different socio-economic backgrounds to have access to quality housing options. This inclusive approach promotes social equity by promoting diverse and economically integrated communities.



The three Fundamental needs that most cities are expected to provide are Housing, Transportation, and Employment. These are the basic standards of living. These three elements are often considered essential for individual well-being and societal stability. Having all three elements fosters Social Equity through inclusivity and accessibility. Transportation plays a significant role in the daily life of citizens; in certain situations, it could be a substantial economic factor, such as a car or a means to job stability. Amsterdam tackles transportation through biking. The city's vast network of traffic-calmed streets and facilities for cyclists, including bike paths, bike racks, and guarded bicycle parking stations, makes cycling convenient and practical for many residents. They also have a system of trams and ferries that can accommodate the less capable of getting around the city. Many factors play a role in the success of inclusivity and accessibility in Amsterdam. Some even argue the way a city's design, characterized by a compact and interconnected urban structure, promotes social interaction and accessibility, its urban design, and green spaces, such as neighborhood parks and open areas, further foster a sense of community and well-being, promoting social equity and cohesion.

In pursuing a just and equitable society, social welfare stands as a cornerstone, representing the collective commitment to ensuring that all individuals, regardless of background or circumstance, have access to essential resources and opportunities that foster a life of dignity, equality, and shared prosperity. Barcelona's superblocks have been instrumental in promoting social welfare and equality by reimagining the city's urban spaces to prioritize people over others. Ildefons Cerdà was responsible for the design of the L'Eixample district in Barcelona, Spain. His innovative approach addressed overcrowding, poor sanitation, and inadequate living conditions in the rapidly growing city. At first, many business owners were upset by this plan, with 5% of their customers coming from cars, but By reclaiming the streets from cars and dedicating them to pedestrians, cyclists, and green spaces, the superblocks have improved air quality, reduced noise pollution, and provided opportunities for recreational activities, contributing to the physical and mental well-being of residents, and an uptick of Bicycles commuters. This prioritization of public space for the community, rather than private vehicles, creates opportunities for cultural events, markets, and social gatherings that celebrate the city's diverse and inclusive character.

The necessity for Environmental Sustainability Continues to be at the forefront of many European cities' challenges to contemporary urban planning. May it be resource depletion, climate change, or the mere impact of industrialization, cities are facing environmental issues in a greater magnitude every year. The rising temperature can pose significant challenges to residents' comfort, health, and well-being due to the lack of widespread air conditioning in milder climate regions. The percentage of CO2 emissions from construction in Europe is generally estimated to be around 36-40% of the EU's total greenhouse gas emissions.

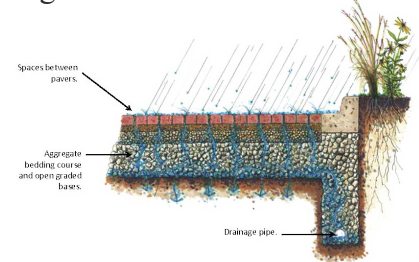


Zurich, Switzerland, is a leader in the battle to tackle climate change; they implemented 2010 "The Seven Principles of Sustainable Construction." The principles provide a framework for building better-quality, affordable homes and infrastructure while reducing construction's carbon footprint and delivering a more sustainable, green built environment. The principles cover sustainable design, durability, energy efficiency, waste reduction, indoor air quality, water conservation, and sustainable building materials. The City of Zurich wants to become climate neutral by 2040; they plan on tackling this through initiatives in Green Building Certifications, Environmental Footprint Reduction, and the 200-watt society model. Zurich Insurance has also launched annual sustainability awards to encourage digitalization and support the reduction of CO2 emissions. Zurich holds 146 LEED (Leadership in Energy and Environmental Design) certifications and 248 BREEAM (Building Research Establishment Environmental).



This reflects the city's commitment to sustainable and environmentally friendly building practices, but it also shows their environmental Footprint Reduction through energy reduction measures and renewable energy initiatives. Switzerland has reduced its emissions of greenhouse gases at an average of 0.6 percent per year since 1990 because of its efforts to push strong policy incentives.

Zurich complements urbanity with various interconnected ecologically valuable habitats, ensuring a high level of biodiversity and a high-quality supply of environmental goods. The Green Book outlines the city's goals and measures for developing green and open spaces, focusing on recreation and nature. Zurich has also been actively involved in sustainable urban development projects, such as Green City, which integrates circular economy principles, sustainable mobility, and renewable energies. These initiatives reflect Zurich's dedication to promoting environmental sustainability and creating a brighter future for its residents.



Athens and Venice, both steeped in history, boast unique urban identities shaped by their distinct origins and architectural styles. Athens, founded in the 5th century BCE, showcases a mix of neoclassical and modern buildings, including the iconic Acropolis. In contrast, Venice, established in the same era, stands out for its water-centric layout, characterized by Byzantine, Gothic, and Renaissance architecture.

The urban layout is a defining feature of both cities. Inland, Athens follows a traditional design with hills, squares, and historical landmarks. Venice, built

on a series of islands connected by canals, is famous for its unique water-centric structure, where canals serve as roads. Economically, Athens is a diverse hub, with sectors like tourism, shipping, and services playing key roles. In comparison, historically a maritime republic, Venice relies heavily on tourism today.

Both cities confront modern challenges. Athens addresses issues like traffic and pollution, prompting sustainable urban planning. Venice deals with rising sea levels and overtourism, leading to initiatives such as the MOSE flood barrier project. Despite their

differences, Athens and Venice are committed to cultural heritage preservation. Athens protects ancient sites and landmarks while balancing development, and Venice focuses on conserving its unique architecture and cultural legacy through restoration projects. These cities exemplify the delicate balance between preserving history and adapting to contemporary urban demands.



A timeline for Athens showing key historical periods: Ancient Athens (500 BCE), Ottoman Era (1458-1833), Independence & Modernization (1833-1920), Post-WWII Reconstruction (1950s-1970s), and Contemporary Development (2000s-Present).



Delving into Malmö's eco-districts unveils a blueprint for sustainable urban development, showcasing innovative initiatives that provide valuable lessons compared to the distinctive urban contexts of Athens and Venice.

Malmö's eco-districts are

characterized by cutting-edge sustainability measures, featuring energy-efficient buildings, green roofs, and an extensive bike infrastructure. These initiatives contribute to a lower carbon footprint and foster a holistic approach to environmentally conscious living. The city's commitment to sustainable urban development is evident in the deliberate design and implementation of these eco-friendly features, creating a model from which other urban centers, including Athens and Venice, can draw inspiration to address their unique challenges.

Exploring Malmö's eco-districts provides essential insights into sustainable urban development, offering valuable lessons compared to the unique characteristics of Athens and Venice.

Sustainable Urban Initiatives:

Malmö's eco-districts exemplify a commitment to environmentally conscious living, featuring energy-efficient buildings, green roofs, and a robust bike infrastructure.

These initiatives contribute to a lower carbon footprint. Similarly, Athens addresses pollution challenges with sustainable strategies, emphasizing public transportation

and green spaces. With its distinctive water-centric layout, Venice focuses on sustainable tourism practices to preserve its environmental integrity amid modern challenges.

Community Engagement and Climate Adaptation: Malmö's eco-districts prioritize community engagement, actively encouraging residents to participate in eco-friendly practices and fostering shared responsibility. This aligns with Athens' emphasis on community involvement in urban planning

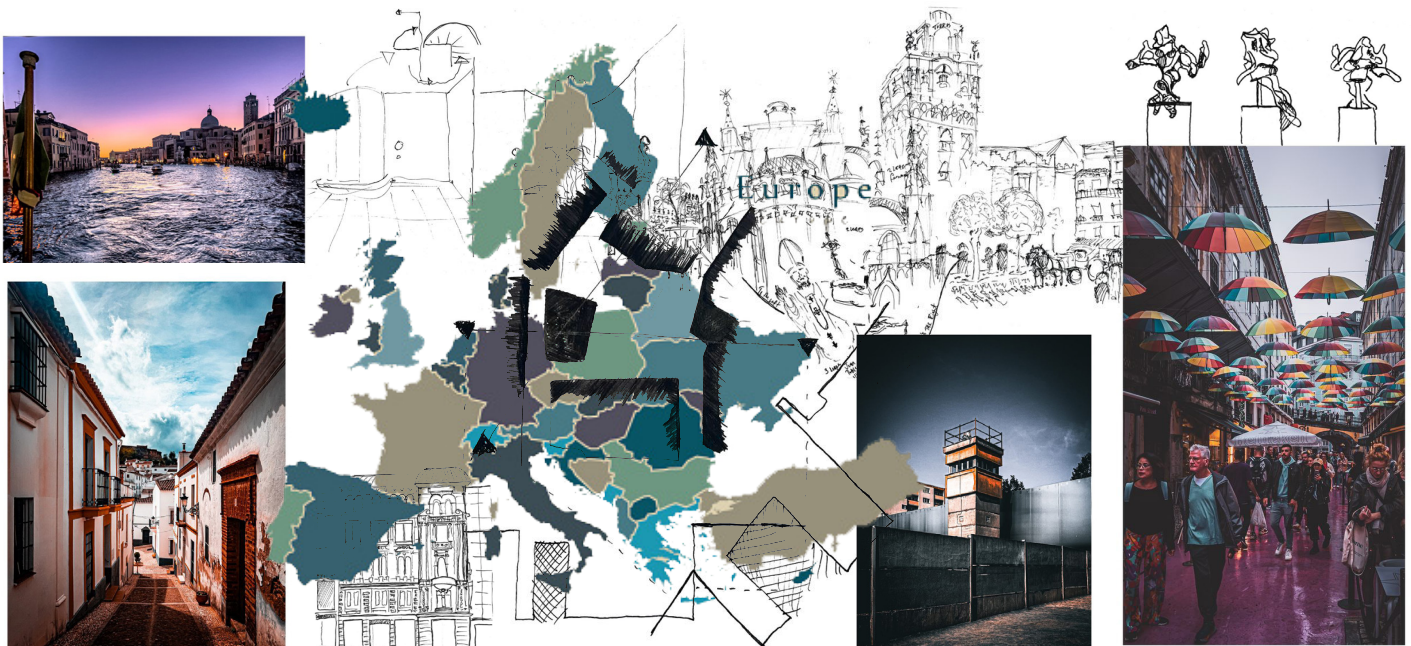
to tackle contemporary issues. Additionally, Malmö's proactive approach to climate adaptation, considering rising sea levels, finds resonance in Athens' focus on resilience through sustainable planning and Venice's implementation of projects like the MOSE flood barrier system.

Economic Diversity and Cultural Preservation: Malmö's eco-districts contribute to economic diversity by creating green jobs in renewable energy and sustainable technologies. This aligns with Athens' diverse economic activities,

promoting green initiatives. Reliant on tourism, Venice explores sustainable practices to diversify its economic landscape. Lessons from Malmö highlight the importance of balancing modern development and cultural preservation. This insight is particularly relevant for Athens and Venice, rich in cultural heritage, encouraging them to adopt sustainable practices while safeguarding their historical assets.

In conclusion, Analyzing the critical determiners of successful urban development using the pillars of sustainability to define success showed us many plausible answers to urban development. To set a list of perfect factors would only lead to fantasy utopia, dystopian societies dreamed up in a scary-fy movie. Instead, success can be defined by the inhabitants. In most cases, success supports and influences the healthy growth of the city and the people within. The healthier living conditions supported by the pillars of sustainability support the facilitation of how urban planning can effectively balance economic growth, social equity, and environmental sustainability. It leads to creating resilient and inclusive urban environments, Which we found in our exploration of Malmos eco-districts and comparing Athens vs. Venice.

Europe is a leader in successful urban development, planning on 80% of its population living in urban centers by 2080. Its relentless history and vast cultures have created a petri dish of modern urban development strategies. Its focus on modernity and the utilization of integration have fostered preservation and innovation, which has led to inclusive and resilient cities it calls home. As the population continues to rise and the environmental issues continue to skyrocket, the lessons learned in the urbanization of Europe will be helpful globally; who knows, one day, the globe may reach the level of integration Europe has achieved.



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